

Relation Algebra

§14. Thinning Algorithms

§14.1. Thinning

(14.1a)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$ (8.1).

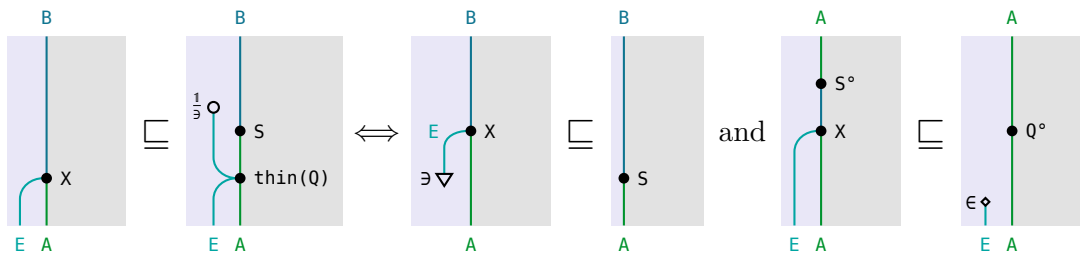
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(14.1b)

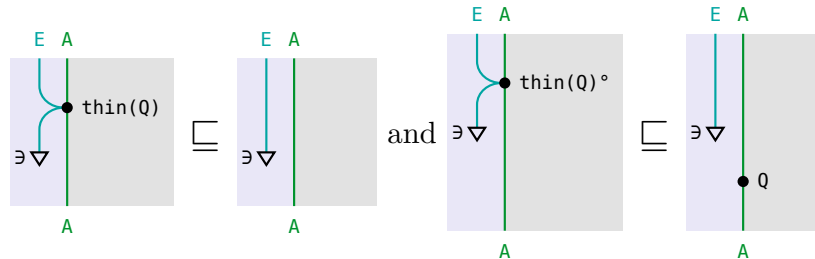
the law	what it says
$X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$	everything kept is an S-value, and every S-value has a Q-lower bound among the kept ones
$\text{thin}(Q) \exists \subseteq \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \subseteq \exists Q$	every element of the set has a Q-lower bound among the kept ones
$Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \subseteq Q \implies \text{thin}(Q) \text{thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \subseteq R \wedge 1 \subseteq Q \wedge R R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (14.1.2a); weaker than the book's “both preorders”: Q transitive is never used and $1 \subseteq R$ follows	thin-introduction : thinning first cannot lose an R-minimum
$\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (14.1.2d)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{union} \subseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§14.1.1. $X \subseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$

(14.1.1a)



(14.1.1b)



(14.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \exists = \wedge(\exists) = 1$

§14.1.2. $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ given $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$

(14.1.2a)

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$

the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

(a) $1 \sqsubseteq \text{thin}(Q)$ — (14.1b)

(14.1.2b)

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$

the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — est-up

(a) $\text{est}(R) \sqsubseteq \exists$ — est-laws

(b) $\text{thin}(Q) \sqsubseteq \exists$ — (14.1.1a)

(14.1.2c)

$\in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$

the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive

(a) $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ — (14.1.1a)

(b) $\in \text{est}(R) \sqsubseteq R^\circ$ — est-laws

(c) R transitive

(14.1.2d)

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$

the singleton holding a Q -least member of a set is a thinning of that set

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \exists$

every member of the singleton is a member of the set

(a) $\triangleq \text{thin}, \exists/\exists$ half

(b) $\frac{1}{\exists} \sqsubseteq 1$ — pow-laws ; $\text{est}(Q) \sqsubseteq \exists$ — est-laws

$\in \text{est}(Q) \frac{1}{\exists} \sqsubseteq Q^\circ \in$

the singleton's member stands in Q to every member of the set

(a) $\triangleq \text{thin}, \in \setminus (Q^\circ \in)$ half

(b) $\in \text{est}(Q) \sqsubseteq Q^\circ$ — est-up

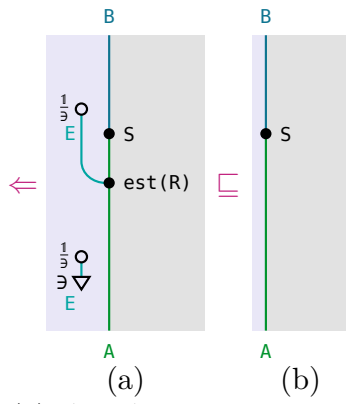
(c) $\frac{1}{\exists} \sqsubseteq \in$ — pow-laws

$$\mathbf{RnS^\circ S \sqsubseteq Q \implies \frac{S}{\exists} \mathbf{est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \mathbf{thin}(Q)}$$

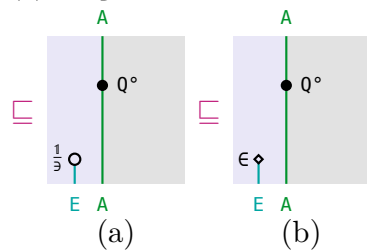
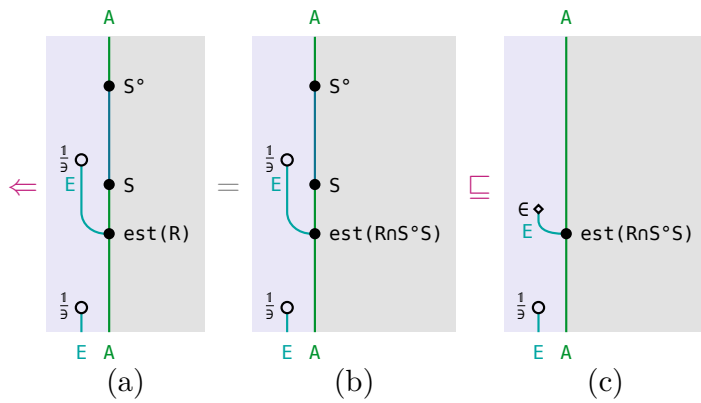
(14.1.2e)

 given $\mathbf{Rn}(S^\circ S) \sqsubseteq Q$, Q a preorder

$$\frac{S}{\exists} \mathbf{est}(R) \frac{1}{\exists} \exists \sqsubseteq S$$

 every member of the singleton is a value S returns


$$\mathbf{RnS^\circ S \sqsubseteq Q \implies S^\circ \frac{S}{\exists} \mathbf{est}(R) \frac{1}{\exists} \sqsubseteq Q^\circ \in$$

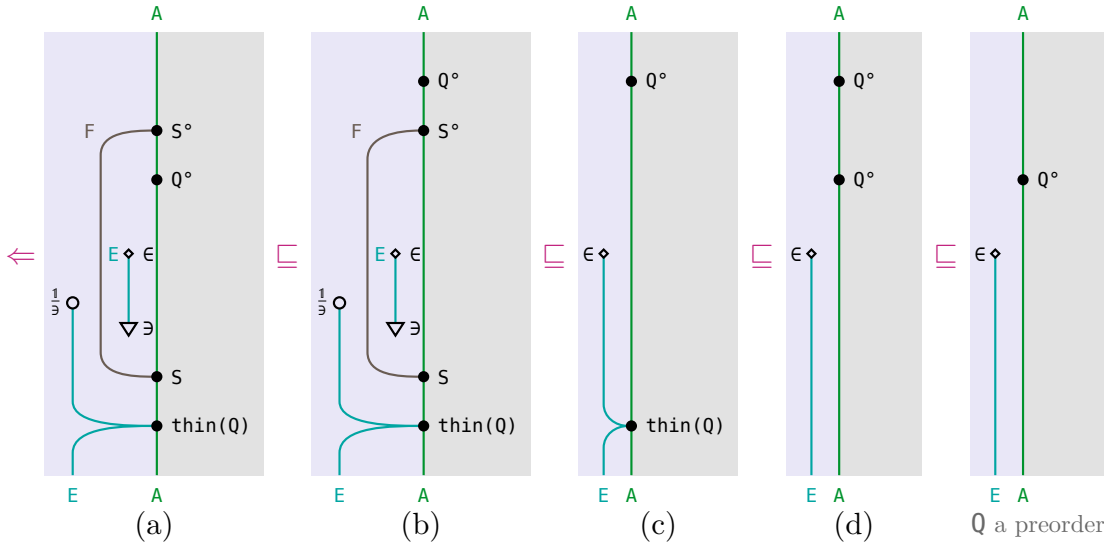
 the singleton's member stands in Q to every value S returns at the same argument


$$QQ \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow (Q \stackrel{F(\exists)S}{\sqsubseteq} \text{thin}(Q)) \sqsubseteq \stackrel{(S)}{\sqsubseteq} \text{thin}(Q) \quad (14.1.2f)$$

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

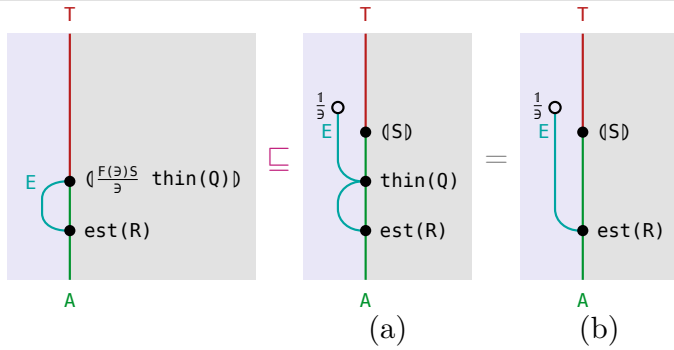
$$(Q \stackrel{F(\exists)S}{\sqsubseteq} \text{thin}(Q)) \sqsubseteq \stackrel{(S)}{\sqsubseteq} \text{thin}(Q) \quad \text{and} \quad (S)^\circ (Q \stackrel{F(\exists)S}{\sqsubseteq} \text{thin}(Q)) \sqsubseteq Q^\circ \in$$

(14.1b) at $X \triangleq (Q \stackrel{F(\exists)S}{\sqsubseteq} \text{thin}(Q))$, (S) for its S



$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge QQ \sqsubseteq Q \wedge RR \sqsubseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow (Q \stackrel{F(\exists)S}{\sqsubseteq} \text{thin}(Q)) \text{est}(R) \sqsubseteq \stackrel{(S)}{\sqsubseteq} \text{est}(R) \quad (14.1.2g)$$

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \sqsubseteq R$, both preorders



§14.2. Paths in a layered network

(14.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.

$\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \pi_2$, $([\text{wrapz}, \text{consw}]) = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

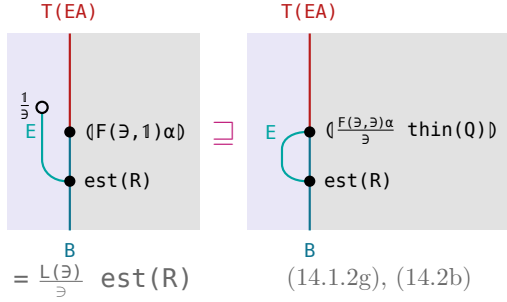
$Q \triangleq R n(\text{head head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{ est}(R)$.

(14.2b)

the law	what it says
$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R n(S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \circ R \circ E \circ \wedge \text{MonoAlg}(F(\exists, 1)\alpha, Q) \wedge Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \quad (14.2c)$$

a least-cost path in a layered network, as a fold over the layers



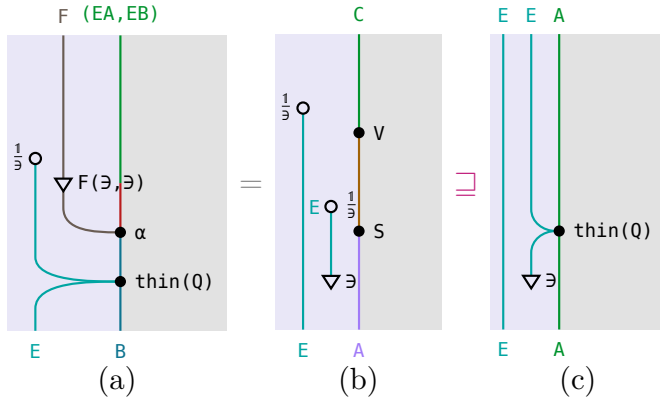
$\sqsubseteq$ (14.2d) under $\downarrow$ $\downarrow$ monotonic: the whole chain runs inside the reduce, and the $\text{est}(R)$ behind it never moves.

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{thin}(Q) \quad (14.2c)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{thin}(Q)$$

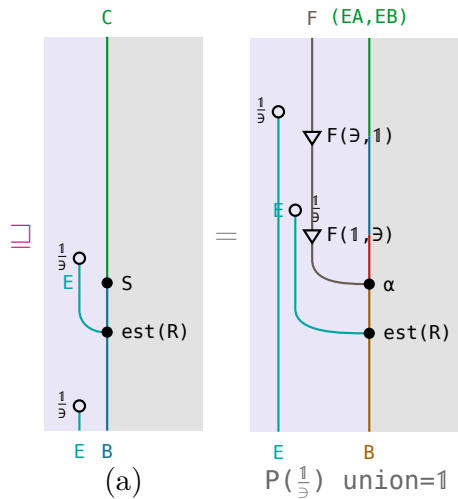
thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



- (a) the algebra of (14.2c) row 2
 (b) $\frac{F(\exists, 1)}{\exists} F(1, \exists)\alpha = \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{union}$
 (c) $\text{union thin}(Q) \sqsubseteq P(\text{thin}(Q)) \text{union} \text{---} (14.1b)$

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

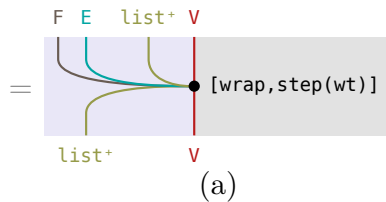
choosing an R-least member of each of those sets is below thinning each and taking the union



- (a) $\frac{F(\exists, 1)}{\exists} \text{thin}(Q) \sqsubseteq \frac{F(\exists, 1)}{\exists} \text{est}(R) \frac{1}{\exists} \text{---} (14.1b), (14.2b)$

$$\text{cp}(F, \text{list}^+(V)) P(\alpha) \text{est}(\text{path}(\text{wt})) = [\text{wrap}, \text{step}(\text{wt})]$$

at the layered network, choosing an R-least path from each case is the program's two cases $[\text{wrap}, \text{step}]$



- (a) α a map; $\frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \text{---} \triangle \text{cost}$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \text{---} \triangle \text{cost}$$

§14.3. Implementing thin

(14.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\langle [\text{nil}, \text{bump } Q] \rangle$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

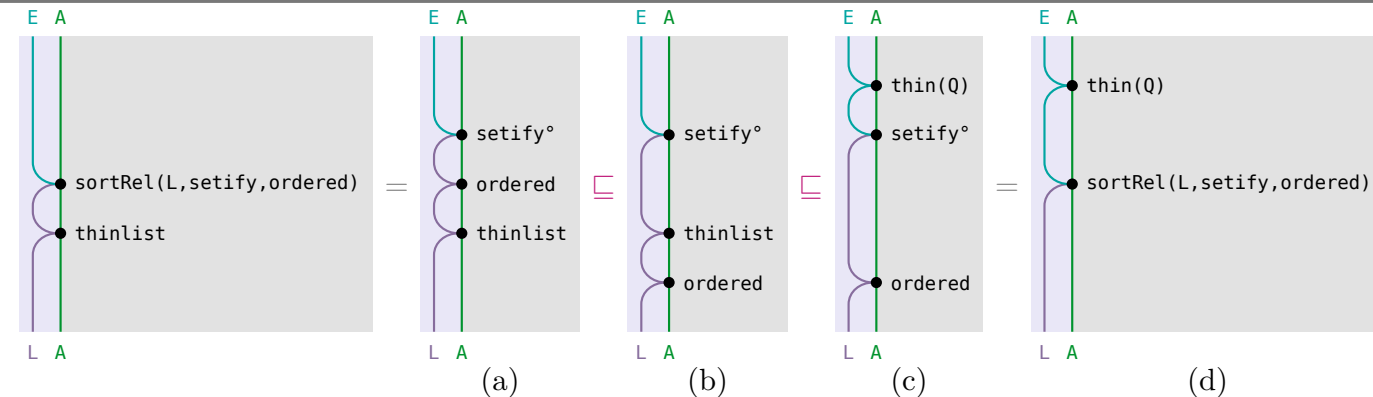
Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

(14.3b)

the law	what it says
$\text{thinlist}(Q) \text{ } xs = [\text{minlist}(Q) \text{ } xs]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (14.3c)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered subseq $\sqsubseteq$ subseq ordered $\wedge$ thinlist setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sortRel(L, setify, ordered) thinlist $\sqsubseteq$ thin(Q) sortRel(L, setify, ordered) (14.3c)

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



- (a) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
 (b) $\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
 (c) $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist}$, dom-laws, triple-chains
 (d) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

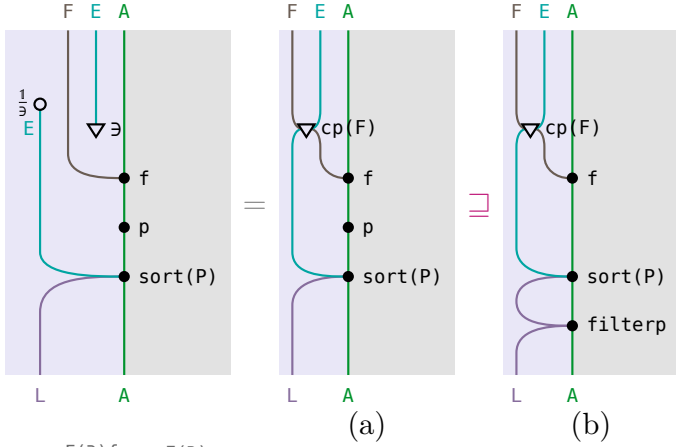
$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \quad (14.3d)$$

$$\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic on P , p coreflexive, F linear.

$$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

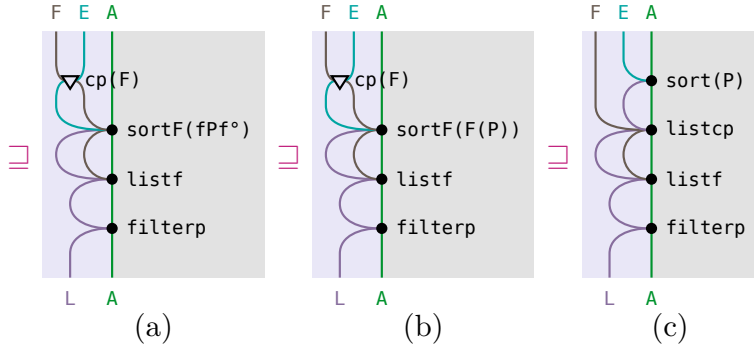
sorting the f -images of the product and filtering by p is below sorting the set $F(\exists)fp$ builds



$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$$

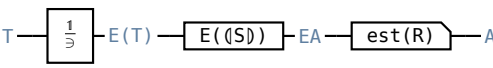
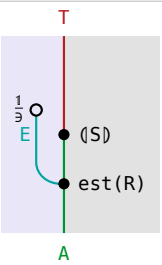
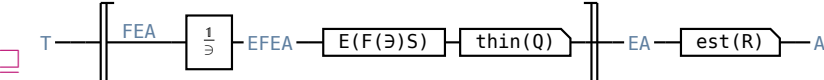
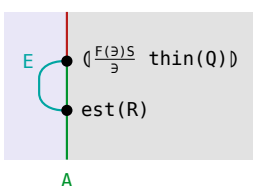
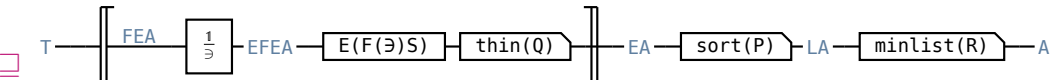
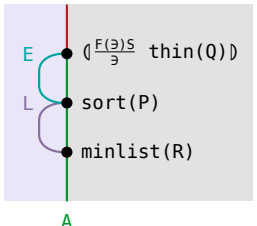
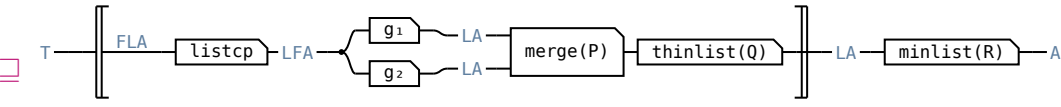
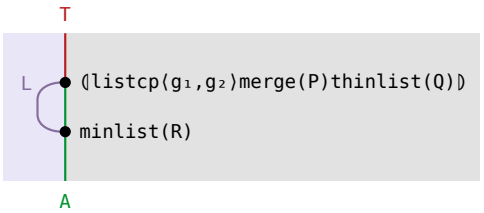
$$F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp}$$

combining the arguments' sorted lists by listcp , mapping f and filtering is below sorting the f -images and filtering



$$\begin{aligned}
 & \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \circ R \circ \sqsubseteq R \circ \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge \text{MonoAlg}(f_2 \text{ } p_2, Q) \wedge (\forall X. \quad (14.3e) \\
 & \quad \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \text{sortF}(f_1 \\
 & \quad P f_1 \circ) \text{listf}_1 \sqsubseteq P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ } P f_2 \circ) \text{listf}_2 \sqsubseteq P(f_2) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_1 \sqsubseteq E(p_1) \text{sort}(P) \\
 & \quad \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq E(p_2) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \wedge \\
 & \quad (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup } \text{sort}(P) \wedge \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \wedge \\
 & \quad \text{sort}(P) \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1 \text{ filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = f_1 \text{ } p_1 \cup f_2 \text{ } p_2 \implies \\
 & \quad (\text{listcp}(g_1, g_2) \text{merge}(P) \text{thinlist}(Q)) \text{minlist}(R) \sqsubseteq \frac{(S)}{\exists} \text{est}(R)
 \end{aligned}$$

a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

circuit	Hinze–Marsden
	
 (14.1.2g) at $f_1 p_1$ and $f_2 p_2$ monotonic on Q — $\triangleq \text{thinlist}$	
 $\text{sort}(P) \text{ minlist}(R) \sqsubseteq \text{est}(R)$ — (14.3b) at its $Q \triangleq R$	
 fusion at (14.3f)	

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (14.3f) \\
& \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \sqsubseteq \text{P}(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \sqsubseteq \text{P}(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \sqsubseteq \text{E}(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq \text{E}(p_2) \text{sort}(P) \wedge \\
& \text{F}(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(\text{F}, \text{A}) \text{sortF}(\text{F}(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \Rightarrow \text{F}(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \\
& \text{merge}(P) \text{thinlist}(Q) \sqsubseteq \frac{\text{F}(\exists)(f_1 \text{ p}_1 \cup f_2 \text{ p}_2)}{\exists} \text{thin}(Q) \text{sort}(P)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit	Hinze–Marsden
$\sqsubseteq$ $\text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \text{ — (14.3b)}$	
$=$ $\{ \frac{\text{F}(\exists)f_1 p_1}{\exists}, \frac{\text{F}(\exists)f_2 p_2}{\exists} \} \text{cup} \text{sort}(P) \text{thinlist}(Q)$ $S = (f_1 p_1) \cup (f_2 p_2) \text{ — } \triangleq \text{thinlist, then } \triangleq \text{cup}$	
$\sqsubseteq$ $\{ \frac{\text{F}(\exists)f_1 p_1}{\exists}, \frac{\text{F}(\exists)f_2 p_2}{\exists} \} (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \text{thinlist}(Q)$ $(\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \text{ — (14.3b)}$	
$=$ $\{ \frac{\text{F}(\exists)f_1 p_1}{\exists} \text{sort}(P), \frac{\text{F}(\exists)f_2 p_2}{\exists} \text{sort}(P) \} \text{merge}(P) \text{thinlist}(Q)$ $\langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{sort}(P), Y \text{sort}(P) \rangle \text{ — bdm-prod-laws}$	
$\sqsubseteq \text{F}(\text{sort}(P)) \text{listcp} \langle g_1, g_2 \rangle \text{merge}(P) \text{thinlist}(Q)$ $(14.3d) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ — bdm-prod-laws; } g_1 \triangleq \text{list}(f_1)$ $\text{filter}(p_1) \text{ — } \triangleq \text{thinlist}$	

§14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2]) \text{ , , within } w \text{ the coreflexive on } xs \text{ with weight } xs \leq w, 0 \leq w.$

$R \triangleq \text{value} \geq \text{value}^\circ, Q \triangleq \text{Rn}(\text{weight} \leq \text{weight}^\circ), P \triangleq R, .$

$\text{FA} = 1 + \text{Item} \times \text{A}, \text{listcp} = \text{wrap} + \text{cpr}, g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1],$
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w), h_2 \triangleq \text{list}(\pi_2).$

(14.4b)

the law	what it says
$(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
$(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$$\begin{aligned}
 & 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\
 & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \\
 & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP } \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sortP} \wedge \\
 & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, [\text{Item}]) \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup } \text{sortP} \wedge \text{sortP} \\
 & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP } \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \\
 & \text{mergeP } \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supseteq} \text{est}(R)
 \end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

(14.4c)

circuit	Hinze–Marsden
$\text{subseq } (\text{within } w) \supseteq \text{est}(R)$	
$\frac{([\text{nil}, \text{cons}](\text{within } w) \cup [\text{nil}, \pi_2])}{\supseteq} \text{est}(R)$ fusion, weights non-negative.	
$(\text{listcp } \langle g_1, g_2 \rangle \text{ merge } R \text{ thinlist}(Q)) \text{minlist}(R)$ binary thinning theorem, at $P \triangleq R$, F linear, Q from (14.4b)	
$\text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{nil}), h_i] \text{ --- } \triangleq \text{within}; \text{minlist}(R)$ is head, packings coming out in descending value	

§14.5. The paragraph problem

(14.5a)

$\text{Line} = \text{list}^+ \text{ Word}$, $\text{Para} = \text{list}^+ \text{ Line}$, $\text{FA} = \text{Word} + \text{Word} \times \text{A}$, $\text{listcp} = \text{wrap} + \text{cpr}$.
 $\text{new}(a, xs) = [[a]] \# xs$, $\text{glue}(a, xs) = [[a] \# \text{head}(xs)] \# \text{tail}(xs)$, $\text{partition} \triangleq ([\text{wrap} \ \text{wrap}, \text{new} \ \cup \ \text{glue}]) : \text{list}^+ \text{ Word} \rightarrow \text{Para}$.
 $\text{width} \triangleq ([\text{length}, (\text{length} \times 1) \text{ plus succ}])$, $0 \leq \text{length } a$, $\text{fits } w$ the coreflexive on a line x with width $x \leq w$, $\text{ok } w$ the coreflexive on $[x] \# xs$ with width $x \leq w$.
 $\text{white } w \ x = w - \text{width } x$, $\text{collect} \triangleq \text{list}(\text{sqr}) \text{ sum}$, $\text{waste } w \triangleq \text{init list}(\text{white } w) \text{ collect}$.
 $R \triangleq (\text{waste } w) \leq (\text{waste } w)^\circ$, $Q \triangleq R \cap (\text{head head}^\circ)$, $P \triangleq \tau$, .
 $g_1 \triangleq \text{list}([\text{wrap} \ \text{wrap}, \text{new}])$, $g_2 \triangleq \text{list}([\text{wrap} \ \text{wrap}, \text{glue}]) \text{ filter}(\text{ok } w)$, $\text{start} \triangleq \text{wrap} \ \text{wrap} \ \text{wrap}$, $h_1 \triangleq \text{list}(\text{new})$, $h_2 \triangleq \text{list}(\text{glue}) \text{ filter}(\text{ok } w)$.

(14.5b)

the law	what it says
$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
$(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$ $(1 \times Q) (\text{glue } (\text{ok } w)) \sqsubseteq \text{glue } (\text{ok } w) Q$ cons monotonic on $\text{collect} \leq \text{collect}^\circ$,	both halves are monotonic on Q once ties in waste are broken by the first line
$\text{merge } \tau = \text{cat}$; $P \triangleq \text{head prefix head}^\circ$ also serves	τ needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

(14.5c)

$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \ \text{wrap}, \text{new}] \tau [\text{wrap} \ \text{wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap} \ \text{wrap}, \text{new}]) \text{sortP} \wedge \text{sortF}([\text{wrap} \ \text{wrap}, \text{glue}] \tau [\text{wrap} \ \text{wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap} \ \text{wrap}, \text{glue}]) \text{sortP} \wedge \text{sortP} \text{ filterp}_2 \sqsubseteq E(\text{ok}(w)) \text{sortP} \wedge$
 $F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, \text{list}^+((\text{list}^+ \text{ Word}))) \text{sortF}(F(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP}$
 $\text{thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}(\text{listf}_1, \text{listf}_2 \text{ filterp}_2)$
 $\text{mergeP} \text{ thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\exists} \text{est}(R)$

a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\frac{\text{list}^+(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^+(\text{Word})) \xrightarrow{E([\text{wrap} \ \text{wrap}, \text{new}] \cup [\text{wrap} \ \text{wrap}, \text{glue}] \text{ok}(w))} E(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^+(\text{list}^+(\text{Word}))}{\text{partition list}^+(\text{fits } w) \xrightarrow{\exists} \text{est}(R)}$	
$= \frac{\text{list}^+(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^+(\text{Word})) \xrightarrow{E([\text{wrap} \ \text{wrap}, \text{new}] \cup [\text{wrap} \ \text{wrap}, \text{glue}] \text{ok}(w))} E(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^+(\text{list}^+(\text{Word}))}{\frac{([\text{wrap} \ \text{wrap}, \text{new}] \cup [\text{wrap} \ \text{wrap}, \text{glue}] \text{ok}(w))}{\exists} \text{est}(R)}$ fusion, every word fits on a line by itself.	
$= \frac{\text{list}^+(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^+(\text{Word})) \xrightarrow{E([\text{wrap} \ \text{wrap}, \text{new}] \cup [\text{wrap} \ \text{wrap}, \text{glue}] \text{ok}(w))} E(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^+(\text{list}^+(\text{Word}))}{\frac{([\text{wrap} \ \text{wrap}, \text{new}] \cup ([\text{wrap} \ \text{wrap}, \text{glue}] \text{ok } w))}{\exists} \text{est}(R)}$ the algebra as $(f_1 p_1) \cup (f_2 p_2)$, $p_1 \triangleq 1 \xrightarrow{\frac{1}{2}} \xrightarrow{\frac{1}{2}} \text{thinlist}$.	
$\frac{([\text{listcp } \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)] \text{minlist}(R)}{\text{binary thinning theorem, at } P \triangleq \tau \text{ with merge } \tau = \text{cat}, Q \text{ from (14.5b)}} \text{minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}$, g_i along the coproduct $\xrightarrow{\frac{1}{2}} \text{partition}$	

§14.6. Bitonic tours

(14.6a)

$FA = (City \times City) + (City \times A)$, the base functor of cons-lists of length at least two; $listcp = wrap + cpr$; $tc : City \times City \rightarrow Real$, neither positive nor symmetric.

$start(a, b) = ([a, b], [a, b])$, $dropl(a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $dropr(a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $tour \triangleq ([start, dropl \cup dropr])$.

$cost(xs, ys) = outcost\ xs + incost\ ys$, $outcost[a_0, \dots, a_n] = tc(a_0, a_1) + \dots + tc(a_{n-1}, a_n)$, $incost[a_0, \dots, a_n] = tc(a_1, a_0) + \dots + tc(a_n, a_{n-1})$.

$next \triangleq tail\ head$, $next2 \triangleq next \times next$, $head2 \triangleq head \times head$, $R \triangleq cost \leq cost^\circ$, $Q \triangleq Rn(next2\ next2^\circ) \cap (head2\ head2^\circ)$, $P \triangleq T$, $g_1 \triangleq list([start, dropl])$, $g_2 \triangleq list([start, dropr])$.

(14.6b)

the law	what it says
$(1 \times R)\ dropl \sqsubseteq dropl\ R$ FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
$(1 \times R)\ dropr \sqsubseteq dropr\ R$ FALSE	
$(1 \times Q)\ dropl \sqsubseteq dropl\ Q$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input
$(1 \times Q)\ dropr \sqsubseteq dropr\ Q$	

(14.6c)

$$\begin{aligned}
 & (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow sortF(X) \sqsubseteq sortF(Y)) \wedge \\
 & sortF([start, droplFn] \tau [start, droplFn]^\circ) g_1 \sqsubseteq P([start, droplFn]) sortP \wedge \\
 & sortF([start, droprFn] \tau [start, droprFn]^\circ) g_2 \sqsubseteq P([start, droprFn]) sortP \wedge \\
 & F(sortP) listcp \sqsubseteq cp(F, dTour(City)) sortF(F \tau) \wedge (sortP \times sortP) cat \sqsubseteq cup\ sortP \wedge sortP \\
 & thinlist \sqsubseteq thin(Qc(tc)) sortP \wedge sortP\ minlist \sqsubseteq est(R) \Rightarrow (listcp(g_1, g_2) cat\ thinlist) minlist \sqsubseteq \\
 & \quad \frac{(tourAlg)}{\exists} est(R)
 \end{aligned}$$

a least-cost bitonic tour, as a fold that thins the tours kept at each city

circuit	Hinze–Marsden
$ \begin{aligned} & dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} E(dCL(City \times City, City)) \xrightarrow{E((tourAlg))} E(Journey(City)^\circ) \xrightarrow{est(R)} Journey(City) \\ & \frac{tour}{\exists} est(R) \end{aligned} $	
$ \begin{aligned} & dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} E(dCL(City \times City, City)) \xrightarrow{E([start, droplFn] \cup [start, droprFn])} E(Journey(City)^\circ) \xrightarrow{est(R)} Journey(City) \\ & = \frac{([start, dropl \cup dropr])}{\exists} est(R) \\ & tour \triangleq ([start, dropl \cup dropr]) \triangleq tour \end{aligned} $	
$ \begin{aligned} & dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} E(dCL(City \times City, City)) \xrightarrow{E([start, droplFn] \cup [start, droprFn])} E(Journey(City)^\circ) \xrightarrow{est(R)} Journey(City) \\ & \sqsubseteq (listcp(g_1, g_2) cat\ thinlist(Q)) minlist(R) \\ & \text{binary thinning theorem, at } P \triangleq T \text{ with merge } \tau = cat, Q \text{ from (14.6b)} \end{aligned} $	
$ \begin{aligned} & ([start\ wrap, cpr\ (list(dropl), list(dropr))\ cat \\ & = thinlist(Q)] minlist(R) \\ & listcp = wrap + cpr, g_i = [list(start), list(drop_i)] \triangleq tour; \\ & \text{quadratic, two tours added per step} \end{aligned} $	